

Course Title: Internet of Things

Semester and Year: Autumn 2024

Credit Structure(L-T-P-Cr): 3-0-2-4

Course Code: IE 407

Prerequisites (if any): Analog Electronics and Digital Logic Design

Instructor's Name with email: VINAY S PALAPARTHY (EMAIL: vinay_shrinivas@daiict.ac.in)

Course Objectives: Hands-on Experience in the IoT Hardware design

Suggested Textbook/references:

- 1) S. Misra, A. Mukherjee, and A. Roy, 2020. Introduction to IoT. Cambridge University Press.
- 2) S. Misra, C. Roy, and A. Mukherjee, 2020. Introduction to Industrial Internet of Things and Industry 4.0. CRC Press.
- 3) Research Papers

Evaluation Scheme

- Labs and Assignments : 40%
- In Sem-1 Exam: 20 %
- In Sem-II Exam: 20 %
- End Sem Exam: 20 %

Lecture Plan : Topic wise lecture plan is expected.

Introduction to IoT system design, Components of system design, Sensors and actuators, Interface electronics, IoT Network topologies, IoT Hardware security design, need of hardware security design for the IoT applications, Hardware design concepts and protocols, Challenges in the hardware security design, Basic implementation of hardware security using the digital devices, Applications for in-situ measurements